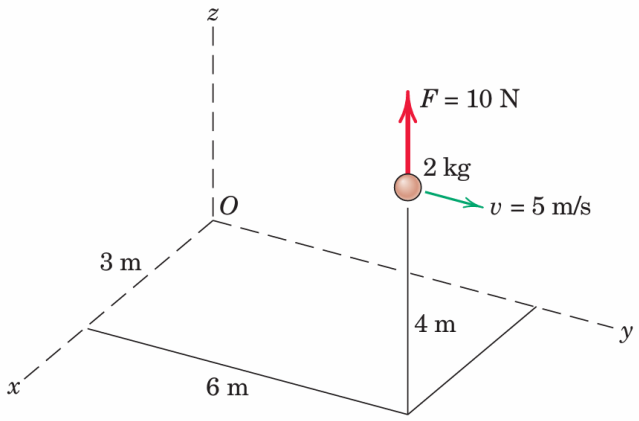
TUTORIAL 6: ANGULAR MOMENTUM AND IMPULSE

ME 1060: DYNAMICS

Question 1

A small sphere has the position and velocity indicated in the figure and is acted upon by the force F. Determine the angular momentum H_O about point O and the time derivative.



Solution. We begin with the definition of angular momentum and write

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$$
$$= (3\mathbf{i} + 6\mathbf{j} + 4\mathbf{k}) \times 2(5\mathbf{j})$$
$$= -40\mathbf{i} + 30\mathbf{k}\ \text{N}\cdot\text{m/s}$$

From Eq. 3/31,

$$\dot{\mathbf{H}}_O = \mathbf{M}_O$$
$$= \mathbf{r} \times \mathbf{F}$$
$$= (3\mathbf{i} + 6\mathbf{j} + 4\mathbf{k}) \times 10\mathbf{k}$$
$$= 60\mathbf{i} - 30\mathbf{j}\ \text{N}\cdot\text{m}$$

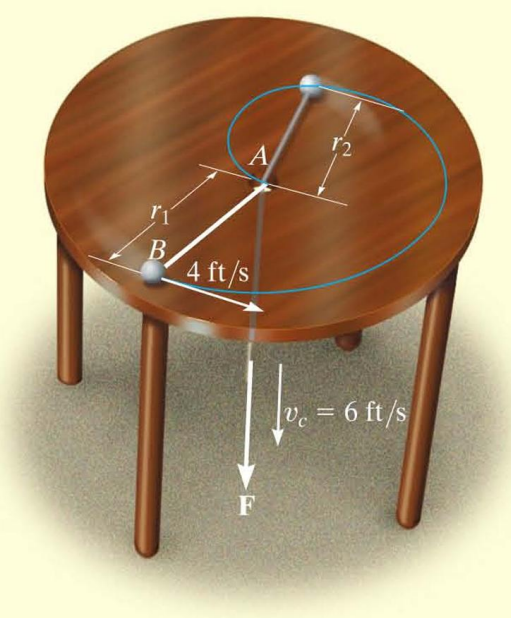
As with moments of forces, the position vector must run from the reference point (O in this case) to the line of action of the linear momentum mv. Here $\boldsymbol{r}$ runs directly to the particle.

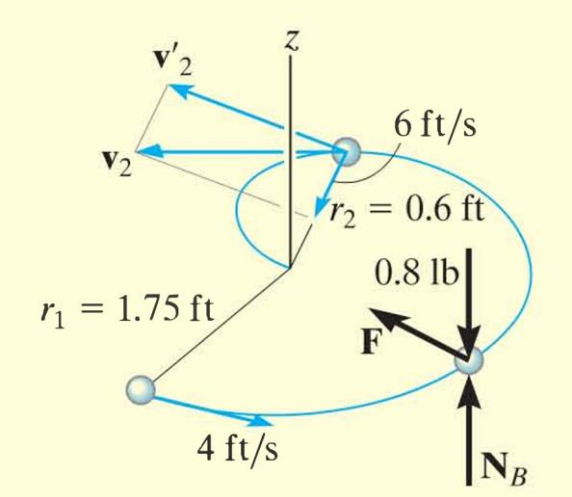
Question 2

The $0.8 - lb$ ball B, shown in figure, is attached to a cord which passes through a hole at A in a smooth table. When the ball is $r_1 = 1.75$ ft from the hole, it is rotating around in a circle such that its speed is $v_1 = 4\ ft/s$. By applying the force F the cord is pulled downward through the hole with a constant speed $v_c = 6\ ft/s$. Determine

(a) the speed of the ball at the instant it is $r_2 = 0.6\ ft$ from the hole, and

(b) the amount of work done by F in shortening the radial distance from r_1 to r_2 . Neglect the size of the ball.





Solution

Part a

Free-Body Diagram.

As the ball moves from r_1 to r_2, the cord force F on the ball always passes through the z axis, and the weight and N_B are parallel to it. Hence the moments, or angular impulses created by these forces, are all zero about this axis. Therefore, angular momentum is conserved about the z axis.

Conservation of Angular Momentum.

The ball's velocity v_2 is resolved into two components. The radial component, $6\ ft/s$, is known; however, it produces zero angular momentum about the z axis. Thus,

$$H_1 = H_2$$

$$r_1 m_B v_1 = r_2 m_B v_2'$$

$$1.75\left(\frac{0.8}{32.2}\right)4 = 0.6\left(\frac{0.8}{32.2}\right)v_2'$$

$$v_2' = 11.67\ ft/s$$

The speed of the ball is thus,

$$v_2 = \sqrt{(11.67)^2 + (6)^2}$$

$$v_2 = 13.1$$

Part b

The only force that does work on the ball is F. (The normal force and weight do not move vertically.) The initial and final kinetic energies of the ball can be determined so that from the principle of work and energy we have,

$$T_1 + \sum U_{1-2} = T_2$$

$$\frac{1}{2}\left(\frac{0.8}{32.2}\right)4^2 + U_F = \frac{1}{2}\left(\frac{0.8}{32.2}\right)13.1^2$$

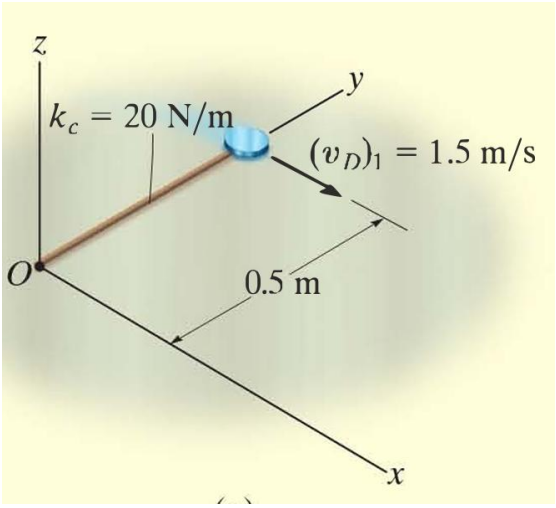
$$U_F = 1.94\ ft.lb$$

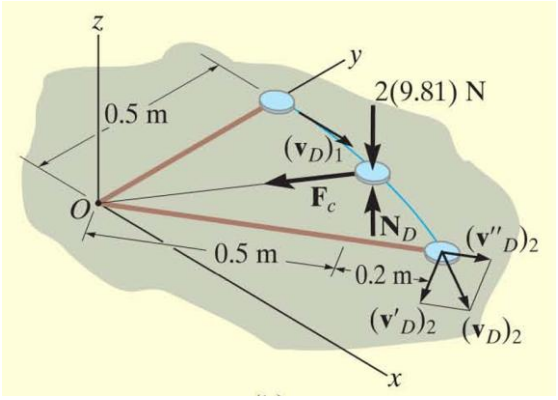
(Gravitation acceleration $g = 32.2 ft/s^2$)

Note: The force F is not constant because the normal component of acceleration, $a_n = v^2/r$ changes as r changes.

Question 3

The 2-kg disk shown in figure rests on a smooth horizontal surface and is attached to an elastic cord that has a stiffness $k_c = 20\ N/m$ and is initially unstretched. If the disk is given a velocity $(v_D)_1 = 1.5\ ms^{-1}$, perpendicular to the cord, determine the rate at which the cord is being stretched and the speed of the disk at the instant the cord is stretched $0.2\ m$.





Free-Body Diagram. After the disk has been launched, it slides along the path shown in Fig. 15–26*b*. By inspection, angular momentum about point O (or the z axis) is *conserved*, since none of the forces produce an angular impulse about this axis. Also, when the distance is 0.7 m, only the transverse component $(\mathbf{v}'_D)_2$ produces angular momentum of the disk about O.

Conservation of Angular Momentum. The component $(\mathbf{v}'_D)_2$ can be obtained by applying the conservation of angular momentum about O (the z axis).

$$(\mathbf{H}_O)_1 = (\mathbf{H}_O)_2$$

$$r_1 m_D (v_D)_1 = r_2 m_D (v'_D)_2$$

$$0.5 \text{ m } (2 \text{ kg})(1.5 \text{ m/s}) = 0.7 \text{ m}(2 \text{ kg})(v'_D)_2$$

$$(v'_D)_2 = 1.071 \text{ m/s}$$

Conservation of Energy. The speed of the disk can be obtained by applying the conservation of energy equation at the point where the disk was launched and at the point where the cord is stretched 0.2 m.

$$T_1 + V_1 = T_2 + V_2$$

$$\tfrac{1}{2} m_D (v_D)_1^2 + \tfrac{1}{2} k x_1^2 = \tfrac{1}{2} m_D (v_D)_2^2 + \tfrac{1}{2} k x_2^2$$

$$\tfrac{1}{2}(2 \text{ kg})(1.5 \text{ m/s})^2 + 0 = \tfrac{1}{2}(2 \text{ kg})(v_D)_2^2 + \tfrac{1}{2}(20 \text{ N/m})(0.2 \text{ m})^2$$

$$(v_D)_2 = 1.360 \text{ m/s} = 1.36 \text{ m/s} \qquad \textit{Ans.}$$

Having determined $(v_D)_2$ and its component $(v'_D)_2$, the rate of stretch of the cord, or radial component, $(v''_D)_2$ is determined from the Pythagorean theorem,

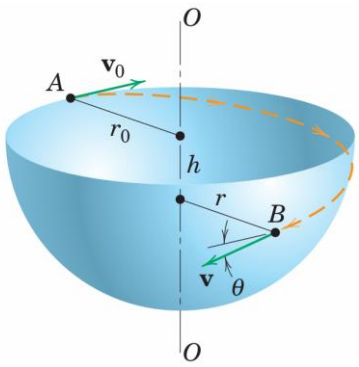
$$(v''_D)_2 = \sqrt{(v_D)_2^2 - (v'_D)_2^2}$$

$$= \sqrt{(1.360 \text{ m/s})^2 - (1.071 \text{ m/s})^2}$$

$$= 0.838 \text{ m/s} \qquad \textit{Ans.}$$

Question 4

A small mass particle is given an initial velocity v_0 tangent to the horizontal rim of a smooth hemispherical bowl at a radius r_0 from the vertical centerline, as shown at point A. As the particle slides past point B, a distance h below A and a distance r from the vertical centerline, its velocity v makes an angle with the horizontal tangent to the bowl through B. Determine θ. (The angle θ is measured in the plane tangent to the hemispherical surface at B).



Solution. The forces on the particle are its weight and the normal reaction exerted by the smooth surface of the bowl. Neither force exerts a moment about the axis O-O, so that angular momentum is conserved about that axis. Thus,

$$[(H_O)_1 = (H_O)_2] \qquad mv_0r_0 = mvr\cos\theta$$

Also, energy is conserved so that $E_1 = E_2$. Thus

$$[T_1 + V_1 = T_2 + V_2] \qquad \frac{1}{2}mv_0{}^2 + mgh = \frac{1}{2}mv^2 + 0$$

$$v = \sqrt{v_0{}^2 + 2gh}$$

Eliminating v and substituting $r^2 = r_0{}^2 - h^2$ give

$$v_0r_0 = \sqrt{v_0{}^2 + 2gh}\sqrt{r_0{}^2 - h^2}\cos\theta$$

$$\theta = \cos^{-1}\frac{1}{\sqrt{1 + \dfrac{2gh}{v_0{}^2}}\sqrt{1 - \dfrac{h^2}{r_0{}^2}}} \qquad \textit{Ans.}$$